

# ECE 501 – DIGITAL IMAGE PROCESSING

## 15-Day Progress Report

### Brain MRI Tumor Segmentation and Quantitative Analysis Using Classical Image Processing

Group 10

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#### 1. Project Objective

The aim of this project is to design a classical digital image-processing pipeline to analyze the brain tumor regions in MRIs. This project deals with classical methodologies which do not use deep learning methods, such as normalization, denoising, contrast enhancement, thresholding, morphological operations, etc.

#### 2. Dataset Selection

The dataset used in this project is the BraTS2020 training & validation dataset. It consists of multi-modal brain MRIs in NIfTI format which include the images of T1, T1Gd/T1ce, T2 and T2-FLAIR, together with their segmentation masks. These ground-truth segmentation masks enable us to evaluate the performance of the classical algorithms quantitatively.

To start with, a subset of 2D slices will be extracted from the whole 3D volume dataset. BraTS2015 2D slices have also been chosen as a simpler dataset.

| Dataset                         | Format               | Role    | Status     |
|---------------------------------|----------------------|---------|------------|
| BraTS2020 Training + Validation | NIfTI (.nii/.nii.gz) | Primary | downloaded |
| BraTS2015 2D slices             | 2D images            | Backup  | Identified |

#### 3. Literature Research

- **Hybrid Multilevel Thresholding Image Segmentation Algorithm in MRI of the Brain:**  
Examines threshold-based image segmentation along with morphological postprocessing.

- **Segmentation of Brain Tumors Using a Clustering Approach With Morphology-Based Technique:** Involves denoising, non-brain tissue removal, clustering, and morphological techniques.
- **Segmentation of Brain Tumors Using Classical Morphology and Discrete Wavelets**  
**Techniques:** Involves the comparison between classical morphology technique and wavelets, involving skull stripping, contrast enhancement and thresholding from histogram.
- **Comparative Analysis of Different Segmentation Techniques Applied to MRI Brain Tumor Images:** Performs comparative analysis of classical techniques including region growing, thresholding, watershed, fuzzy C-means and histogram thresholding.

#### 4. 15-Days Progress Activities

- **Days 1–3: Dataset and data loading**

Organize BraTS2020, load NIfTI volumes using Python/NiBabel, examine dimensionality, take 2D axial slices and find slices that contain tumors based on the segmentation mask.

- **Days 4–6: Preprocessing**

Study the effects of intensity normalization, basic Gaussian/median denoising and contrast enhancement via histogram equalization/CLAHE.

- **Days 7–9: Skull stripping**

Formulate the first version of classical brain-mask extraction procedure that involves thresholding, morphological operations and connected components and form skull-stripped MRI images.

- **Days 10–12: Baseline tumor segmentation**

Apply thresholding as the basic approach and perform the morphological postprocessing for cleaning the tumor mask.

- **Days 13–15: Evaluation and documentation**

Compare results against the ground truth using Dice/IoU, prepare the visualization, document failure cases and start the Introduction/Methodology sections.

## **5. Proposed Classical Processing Pipeline**

1. BraTS2020 NIfTI MRI volume
2. The next step is to choose a slice from a 2D volume that contains a tumor.
3. Intensity normalization
4. Denoising
5. Contrast enhancement
6. Brain/skull-region extraction
7. Segmentation of tumors using thresholds as a baseline.
8. Morphological refinement
9. Predicted tumor mask
10. Comparison with ground-truth mask
11. Dice and IoU evaluation

## **6. Expected Evidence for the Next Progress Check**

- Successful loading and visualization of BraTS2020 MRI volumes.
- Sample T1, T1ce, T2 and FLAIR slices, with FLAIR being done first as a baseline experiment.
- Corresponding ground-truth tumor masks.
- Examples of preprocessed MRI images with denoising and contrast enhancement.
- An initial skull stripped MRI output.
- First time tumor segmentation mask based on threshold cut-off.
- Visual comparison between the original MRI, ground truth and predicted mask.
- Initial dice & IoU results for small number of patients/slices.
- Use examples of successful and unsuccessful segmentation to explain how segmentation works.

## **7. Planned Future Extensions**

The first 15 days are for setting the baseline that is to be used, but not for completing the entire project. The next phases will include analysis of classical segmentation techniques, analysis of robustness of the proposed method against noise, blur and contrast degradation, and finally, morphological analysis of tumor segment area.

## **8. Current Challenges**

BraTS data are 3D NIfTI volumes, requiring suitable 2D slice selection for the initial experiments. The intensity distributions of MRI differ from modality to patient, and care must be taken in preprocessing and normalizing them. Thresholding might result in false positive regions, requiring morphological post processing and comparison with ground truth.

## **9. Summary**

The project has created the dataset and the direction of the literature and initial classical image-processing pipeline. The primary aim is to get a working baseline from BraTS2020 FLAIR images and their ground truth masks. This baseline will be used for later method comparison, robustness testing and quantitative tumor morphology analysis.